

Topic 2.10 - Inverses of Exponential Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Explain logarithms as inverse functions of exponentials.
- Connect inverse pairs through compositions, points, graphs, domains, ranges, and asymptotes.
- Construct inverses of basic and transformed exponential functions.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Simplify: (a) $\log_7(7^{x+4})$; (b) $7^{\log_7(2x-5)}$ with its restriction.

Question 2

Core skill

No calculator

If $f(x) = 3^x$, list three corresponding points on f^{-1} using $x = -1, 0, 2$ on f .

Question 3

Core skill

No calculator

Find the inverse of $f(x) = 5^x + 8$ and state its domain, range, and vertical asymptote.

Question 4

Core skill

No calculator

Find the inverse of $g(x) = 2(4)^{x+3} - 7$.

Question 5

Core skill

No calculator

Find the inverse of $h(x) = 9 - 6(2)^x$ and state the inverse domain.

Question 6

Core skill

No calculator

Verify that $f(x) = 2^x - 1$ and $g(x) = \log_2(x + 1)$ are inverse functions.

Section B: Application, comparison, and error analysis

Question 7

Context

Calculator permitted

The exponential $P(t) = 500(1.12)^t$ maps time to population. Find the inverse and interpret $P^{-1}(1000)$.

Question 8

Structure

No calculator

An exponential has horizontal asymptote $y = -4$ and range $(-4, \infty)$. What are the inverse domain and vertical asymptote?

Question 9

Error analysis

No calculator

A student claims the inverse of $2^x + 5$ is $2^{-x} - 5$. Explain the error.

Question 10

Parameter reasoning

No calculator

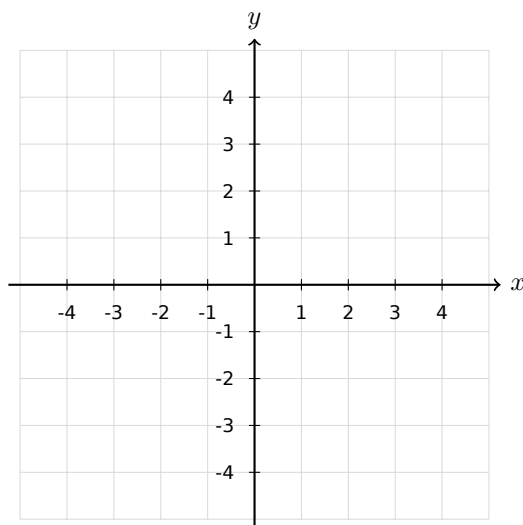
Find a and k if an exponential $f(x) = a(3)^x + k$ has inverse domain $(2, \infty)$ and $f(0) = 7$, assuming $a > 0$.

Question 11

Graphing

No calculator

Sketch $f(x) = 2^x + 1$ and its inverse on the same axes. Label both asymptotes and two reflected point pairs.

**Question 12**

Inverse reasoning

No calculator

Solve $f^{-1}(x) = 3$ without first writing a formula if $f(x) = 4(2)^{x-1} + 6$.

Section C: AP-style multiple choice**MCQ 1**

No calculator

The inverse of $f(x) = 3^x + 4$ is:

- A. $3^{-x} - 4$
- B. $\log_3(x) + 4$
- C. $\log_3(x - 4)$
- D. $\log_3(4 - x)$

MCQ 2

No calculator

If the horizontal asymptote of an exponential is $y = -2$, the vertical asymptote of its inverse is:

- A. $x = -2$
- B. $y = -2$
- C. $x = 2$
- D. $y = 2$

MCQ 3

No calculator

Which identity is valid for $x > 0$?

- A. $\log_b x = b^x$
- B. $b^{\log_b x} = x$
- C. $\log_b(b + x) = x$
- D. $b \log_b x = x$

MCQ 4

No calculator

If $(2, 17)$ lies on f , which point lies on f^{-1} ?

- A. $(-2, -17)$
- B. $(17, 2)$
- C. $(-17, -2)$
- D. $(2, 17)$ only

Section D: Free-response reasoning

FRQ 1 - Transformed inverse

No calculator

6 points

Let $f(x) = 8(3)^{x-2} - 5$. (a) Find $f^{-1}(x)$. (b) State the domain, range, and vertical asymptote of f^{-1} . (c) Verify $f^{-1}(f(x)) = x$. (d) Identify the inverse point corresponding to the anchor point of f .

FRQ 2 - Inverse model in context

Calculator permitted

6 points

A fund is modeled by $B(t) = 12000(1.045)^t$, where t is years and B is QAR. (a) Find B^{-1} and state its domain. (b) Interpret the inverse function. (c) Estimate when the balance reaches 18000 QAR. (d) Explain one limitation of interpreting $B^{-1}(y)$ for every $y > 0$.
